Product info

KonstruX ST ZK 6,5x140		**Number of screws per joint 2**	
Article number 904811	100 Piece	Diameter screw / thread	6.5 mm
Length	140 mm	Diameter head	8.0 mm
Length screw head	5.5 mm	Diameter core	4.5 mm

User input

Side flaps

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled
Width = 100 mm | Height = 120 mm | Overlap = 250 mm

Carrier

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled
Width = 160 mm | Height = 240 mm

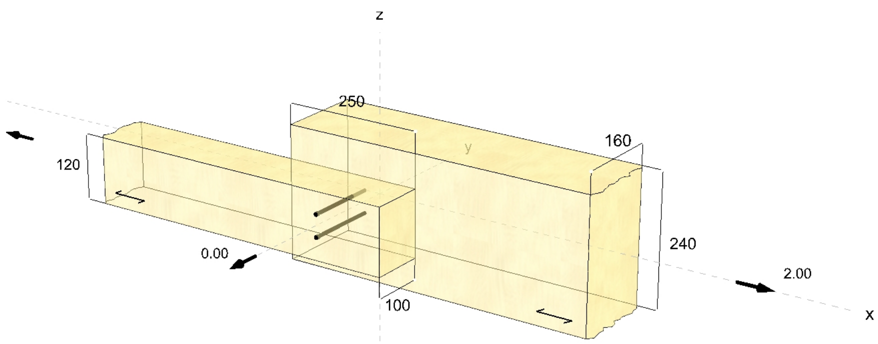
Load

Class of usage: 1

Design loads
x-direction = 2.00 kN | Duration of load: medium

Screws

Configuration: straight | Screwing: flush beam



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Screw distances

Screwing angle	90 °
Drilling depth	0 mm

switched indices

Distances - Side flaps [mm]	Minimum	Existing	
$a_{3\,c}$	98 /6.5=15	125	EN 1995-1-1
$a_{4\,t}$	33 /6.5=5	40	EN 1995-1-1
a_2	33	40	EN 1995-1-1

Distances - Carrier [mm]	Minimum	Existing	
$a_{3,c}$	98	125	EN 1995-1-1
$a_{4,t}$	33	100	EN 1995-1-1
a_2	33	40	EN 1995-1-1

- Which Eurocode annex?
- Why treated as nail, not bolt?
- Which timber density used?
- Same result for 8mm screws
- Screw is not considered self drilling?

8.7 Screwed connections

8.7.1 Laterally loaded screws

(1)P The effect of the threaded part of the screw shall be taken into account in determining the load-carrying capacity, by using an effective diameter d_{ef}.

(2) For smooth shank screws, where the outer thread diameter is equal to the shank diameter, the rules given in 8.2 apply, provided that:
- The effective diameter d_{ef} is taken as the smooth shank diameter;
- The smooth shank penetrates into the member containing the point of the screw by not less than $4d$.

(3) Where the conditions in (2) are not satisfied, the screw load-carrying capacity should be calculated using an effective diameter d_{ef} taken as 1,1 times the thread root diameter.

(4) For smooth shank screws with a diameter $d > 6$ mm, the rules in 8.5.1 apply.

(5) For smooth shank screws with a diameter of 6 mm or less, the rules of 8.3.1 apply.

(6) Requirements for structural detailing and control of screwed joints are given in 10.4.5.

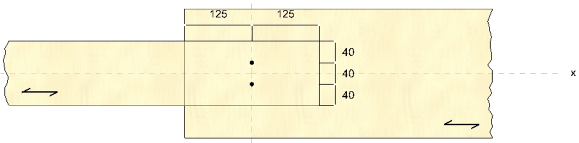

Table 8.2 – Minimum spacings and edge and end distances for nails

Spacing or distance (see Figure 8.7)	Angle α	Minimum spacing or end/edge distance		
		without predrilled holes		with predrilled holes
		$\rho_k \leq 420$ kg/m³	420 kg/m³ $< \rho_k \leq 500$ kg/m³	
Spacing a_1 (parallel to grain)	$0^\circ \leq \alpha \leq 360^\circ$	$d < 5$ mm: $(5+5\lvert\cos\alpha\rvert)\,d$ $d \geq 5$ mm: $(5+7\lvert\cos\alpha\rvert)\,d$	$(7+8\lvert\cos\alpha\rvert)\,d$	$(4+\lvert\cos\alpha\rvert)\,d$
Spacing a_2 (perpendicular to grain)	$0^\circ \leq \alpha \leq 360^\circ$	$5d$	$7d$	$(3+\lvert\sin\alpha\rvert)\,d$
Distance $a_{3,t}$ (loaded end)	$-90^\circ \leq \alpha \leq 90^\circ$	$(10+5\cos\alpha)\,d$	$(15+5\cos\alpha)\,d$	$(7+5\cos\alpha)\,d$
Distance $a_{3,c}$ (unloaded end)	$90^\circ \leq \alpha \leq 270^\circ$	$10d$	$15d$	$7d$
Distance $a_{4,t}$ (loaded edge)	$0^\circ \leq \alpha \leq 180^\circ$	$d < 5$ mm: $(5+2\sin\alpha)\,d$ $d \geq 5$ mm: $(5+5\sin\alpha)\,d$	$d < 5$ mm: $(7+2\sin\alpha)\,d$ $d \geq 5$ mm: $(7+5\sin\alpha)\,d$	$d < 5$ mm: $(3+2\sin\alpha)\,d$ $d \geq 5$ mm: $(3+4\sin\alpha)\,d$
Distance $a_{4,c}$ (unloaded edge)	$180^\circ \leq \alpha \leq 360^\circ$	$5d$	$7d$	$3d$

Table 8.4 – Minimum values of spacing and edge and end distances for bolts

Spacing and end/edge distances (see Figure 8.7)	Angle	Minimum spacing or distance
a_1 (parallel to grain)	$0^\circ \leq \alpha \leq 360^\circ$	$(4+\lvert\cos\alpha\rvert)\,d$
a_2 (perpendicular to grain)	$0^\circ \leq \alpha \leq 360^\circ$	$4\,d$
$a_{3,t}$ (loaded end)	$-90^\circ \leq \alpha \leq 90^\circ$	max $(7\,d$; 80 mm)
$a_{3,c}$ (unloaded end)	$90^\circ \leq \alpha < 150^\circ$ $150^\circ \leq \alpha < 210^\circ$ $210^\circ \leq \alpha \leq 270^\circ$	A1 $(1+6\sin\alpha)\,d$ $4\,d$ $(1+6\lvert\sin\alpha\rvert)\,d$ A1
$a_{4,t}$ (loaded edge)	$0^\circ \leq \alpha \leq 180^\circ$	max $[(2+2\sin\alpha)\,d$; $3d]$
$a_{4,c}$ (unloaded edge)	$180^\circ \leq \alpha \leq 360^\circ$	$3\,d$

Thread in the side flaps

$n = 1$

$n_{ef} = n^{0.9} = 1.00$ EN 1995-1-1 8.7.2 (8) (8.41)

$\alpha = 90\,°$ ETA-11/0024

$k_{ax} = 1.0$ ETA-11/0024

$f_{ax,k} = 11.40\,\frac{N}{mm^2}$ ETA-11/0024

$d = 6.5\,mm$ ETA-11/0024

$l_{ef} = 95\,mm$

$\rho_k = 350\,\frac{kg}{m^3}$ EN 338 5 EN 14080 5.1.4.3 (4)(5)

$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350}\right)^{0,8} = 7.00\,kN$ ETA-11/0024

Thread in the support

$n = 1$

$n_{ef} = n^{0.9} = 1.00$ EN 1995-1-1 8.7.2 (8) (8.41)

$\alpha = 90\,°$ ETA-11/0024

$k_{ax} = 1.0$ ETA-11/0024

$f_{ax,k} = 11.40\,\frac{N}{mm^2}$ ETA-11/0024

$d = 6.5\,mm$ ETA-11/0024

$l_{ef} = 36\,mm$

$\rho_k = 350\,\frac{kg}{m^3}$ EN 338 5 EN 14080 5.1.4.3 (4)(5)

$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350}\right)^{0,8} = 2.65\,kN$ ETA-11/0024

Tensile load capacity

$n = 1$

$n_{ef} = \mathrm{n}^{0.9} = 1.00$ EN 1995-1-1 8.7.2 (8) (8.41)

$f_{tens,k} = 17.00\,kN$ ETA-11/0024

$F_{t,Rk} = n_{ef} \cdot f_{tens,k} = 17.00\,kN$ EN 1995-1-1 8.7.2 (7) (8.40c)

Shear

$V_{d,S} = 2.00\,kN$

$k_{mod,1} = 0.80$ EN 1995-1-1 3.1.3 (1)

$k_{mod,2} = 0.80$ EN 1995-1-1 3.1.3 (1)

$k_{mod} = \sqrt{k_{mod,1}k_{mod,2}} = 0.80$ EN 1995-1-1 2.3.3.1 (2) (2.6)

$n_{0,1} = 1$

$n_{ef,0,1} = 1.00$ EN 1995-1-1 8.3.1.1 (8) (8.17)

$n_{90,1} = 2$

$n_{0,2} = 1$

$n_{ef,0,2} = 1.00$ EN 1995-1-1 8.3.1.1 (8) (8.17)

$n_{90,2} = 2$

$n = min\left(n_{ef,0,1} \cdot n_{90,1}\,; n_{ef,0,2} \cdot n_{90,2}\right) = 2.00$

$\rho_{k,1} = 350\,\frac{kg}{m^3}$ EN 338 5 EN 14080 5.1.4.3 (4)(5)

$\rho_{k,2} = 350\,\frac{kg}{m^3}$ EN 338 5 EN 14080 5.1.4.3 (4)(5)

$f_{h,1,k} = 16.37\,\frac{N}{mm^2}$ ETA-11/0024

$f_{h,2,k} = 16.37\,\frac{N}{mm^2}$ ETA-11/0024

$t_1 = 100\,mm$ EN 1995-1-1 8.2.2 (1)

$t_2 = 40\,mm$ EN 1995-1-1 8.2.2 (1)

$\beta = \frac{f_{h,2,k}}{f_{h,1,k}} = 1.00$ EN 1995-1-1 8.2.2 (1) (8.8)

$M_{y,k} = 15.0\,Nm$ ETA-11/0024

$F_{ax,Rk} = 2.65\,kN$ EN 1995-1-1 8.2.2 (2)

(a) $f_{h,1,k}t_1 d = 10.64\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(b) $f_{h,2,k}t_2 d = 4.26\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(c) $\frac{f_{h,1,k}t_1 d}{1+\beta}\left[\sqrt{\beta+2\beta^2\left[1+\frac{t_2}{t_1}+\left(\frac{t_2}{t_1}\right)^2\right]+\beta^3\left(\frac{t_2}{t_1}\right)^2}-\beta\left(1+\frac{t_2}{t_1}\right)\right]+\frac{F_{ax,Rk}}{4} = 4.22\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(d) $1{,}05\,\frac{f_{h,1,k}t_1 d}{2+\beta}\left[\sqrt{2\beta(1+\beta)+\frac{4\beta(2+\beta)M_{y,k}}{f_{h,1,k}dt_1^2}}-\beta\right]+\frac{F_{ax,Rk}}{4} = 4.54\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(e) $1{,}05\,\frac{f_{h,1,k}t_2 d}{1+2\beta}\left[\sqrt{2\beta^2(1+\beta)+\frac{4\beta(1+2\beta)M_{y,k}}{f_{h,1,k}dt_2^2}}-\beta\right]+\frac{F_{ax,Rk}}{4} = 2.52\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(f) $1{,}15\sqrt{\frac{2\beta}{1+\beta}}\sqrt{2M_{y,k}f_{h,1,k}d}+\frac{F_{ax,Rk}}{4} = 2.72\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

$F_{v,Rk} = min\left(F_{v,Rk}(a);F_{v,Rk}(b);F_{v,Rk}(c);F_{v,Rk}(d);F_{v,Rk}(e);F_{v,Rk}(f)\right) = 2.52\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

$\gamma_M = 1.30$ ÖNORM B 1995-1-1 2.4.1(1) P

$$F_{v,Rd} = k_{mod} \cdot \frac{F_{v,Rk}}{\gamma_M} = 1.55\,kN$$

EN 1995-1-1
2.4.3 (1)P (2.17)

$$\eta = \left(\frac{V_{d,S}}{n \cdot F_{v,Rd}} \right) \cdot 100\,\% = 64.42\,\%$$

The system has been calculated successfully

Hints

- The calculation is according to:
- EN 338 (2016-07) + EN 14081-1 (2016-06) EN 14080 (2013-09)
- EN 1990 (2010-12)
- EN 1991-1-1 (2010-12)
- EN 1993-1-1 (2010-12)
- EN 1995-1-1 (2010-12) + EN 1995-1-1/A2 (2014-07) + ÖNORM B 1995-1-1 (2015-06)
- DIN EN 1995-1-1/NA (2013-08)
- ETA-11/0024 (2017-03)
- The screws may only be used when subjected predominantly to dead loads.
- Verification of the components must be carried out separately if necessary.
- Attention: Please check the assumptions. The displayed values, types and numbers of fasteners are dimensioning aids only.
  Projects can only be dimensioned by authorized people.
  If you need a verification of stability please contact an authorized civil engineer. We can gladly help you with respective contacts.